Q1. (a) In producing high yields of cereal many farmers use very large machinery and rely heavily on fertilisers and pesticides.

Describe **three** ways in which these agricultural methods can damage ecosystems.

1. .....

.....

2. .....

.....

3. .....

.....

3 marks

(b) Intensively reared cattle eat grain to produce animal protein in milk and meat. For every ten kilojoules of energy in the cereals grown as cattle food, about one kilojoule is stored in cattle.

Give **two** reasons why the amount of energy stored in cattle is so much lower than the amount of energy in the cereal crop.

1. .....

.....

2. .....

.....

2 marks

Maximum 5 marks

Q1.

(a) any **three** from

- fertilisers cause eutrophication

  accept a description of eutrophication, such as 'fertilisers cause growth of algae, algae die and are decomposed lowering the oxygen concentration'

  accept 'fertilisers around the fields encourage fast-growing plants and reduce variety'

- pesticides destroy food chains
- monocultures allow diseases to spread quickly
- monocultures restrict **or** reduce the food supply for birds
- habitats are destroyed **or** hedgerows are lost

  accept 'machinery crushes plants'

- heavy machinery damages soil structure

  *accept 'they cause soil erosion' **or** 'machinery compacts soil'*

  *accept 'pollution from machinery poisons plants' **or** 'exhaust fumes kill plants'*

  *accept 'pollution from exhaust fumes **or** fertilisers **or** pesticides **or** nitrates' as an alternative to no more than one point*

  *do **not** accept 'pollution'*

3

(b) any **two** from

- they use energy to keep warm **or** they lose heat
- they use energy to move
- they lose energy **or** heat in urine
- they lose energy **or** heat in faeces **or** not all the food is digested
- not all the cereal crop is eaten

2

[5]